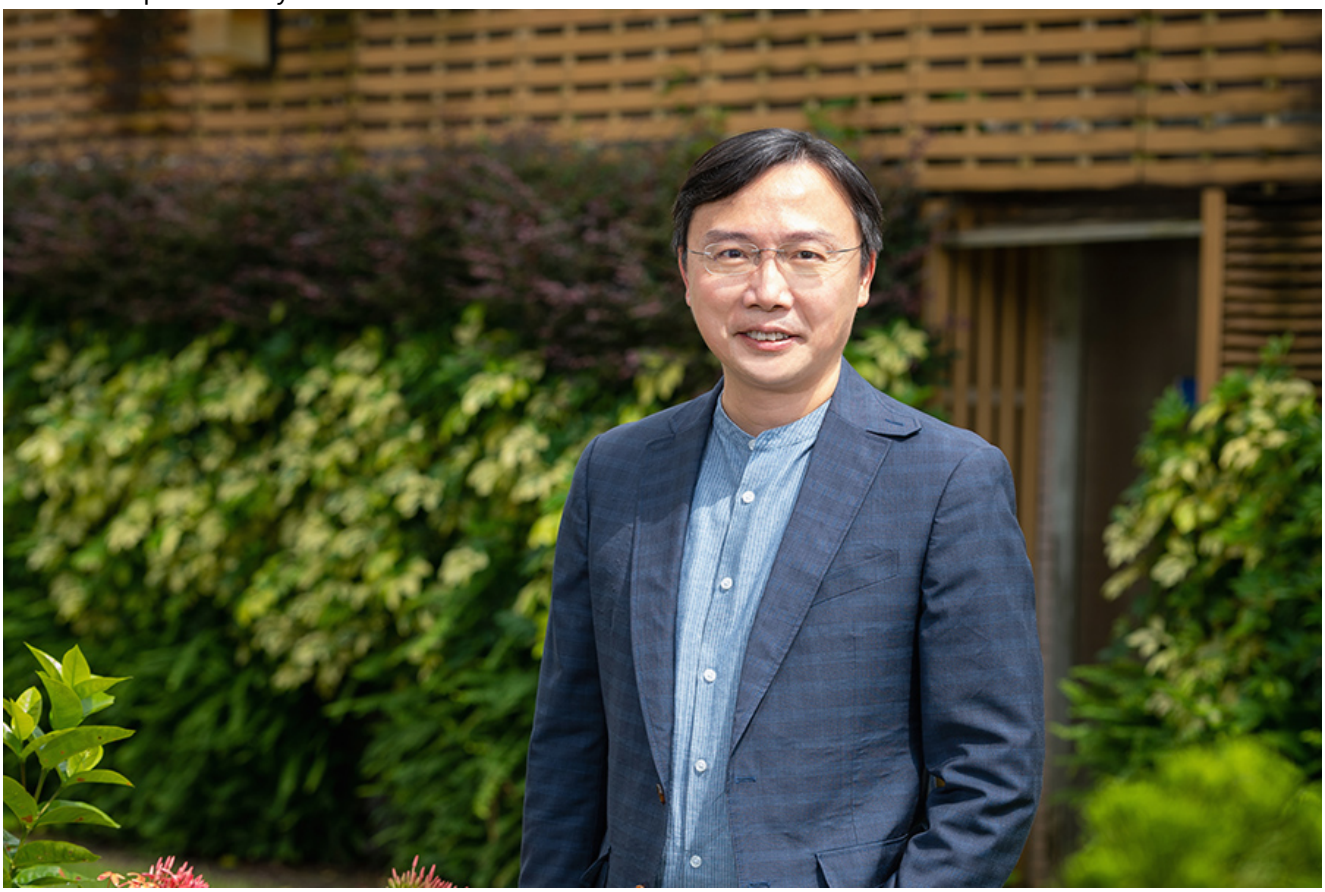

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About

William K. Cheung received the Ph.D. degree in computer science from the Hong Kong University of Science and Technology in Hong Kong in 1999. He is currently the Professor of the Department of Computer Science, Hong Kong Baptist University, Hong Kong. His current research interests include artificial intelligence, data mining, collaborative information filtering, social network analysis, and healthcare informatics. He has served as the Co-Chairs and Program Committee Members for a number of international conferences on areas including artificial intelligence, machine learning, data mining, Web intelligence, and health informatics. He is the Program Co-chair of 2020 IEEE International Conference on Health Informatics. From 2002-2018, he was on the Editorial Board of the IEEE Intelligent Informatics Bulletin. He is a Track Editor of Web Intelligence Journal and an Associate Editor of Journal of Health Information Research.

Research Interests

- Artificial Intelligence
- Data Mining
- Social Network Analysis
- Health Informatics

Selected Publications

- K. Yin, W.K. Cheung, B.C.M. Fung and J. Poon, “Learning Inter-Modal Correspondence and Phenotypes from Multi-Modal Electronic Health Records,” in *IEEE Transactions on Knowledge and Data Engineering (to appear), 2020*
- L. Liu, X. Li, W. K. Cheung and L. Liao, "Structural Representation Learning for User Alignment Across Social Networks," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 32, no. 9, pp. 1824-1837, 2020
- K. Yin, A. Afshar, J.C. Ho, W.K. Cheung, C. Zhang, J. Sun, “LogPar: Logistic PARAFAC2 Factorization for Temporal Binary Data with Missing Values,” in *Proceedings of the 26th SIGKDD Conference on Knowledge Discovery and Data Mining*, August 22 - 27, 2020, San Diego, CA – USA
- D. Qian and W.K. Cheung, “Enhancing Variational Autoencoders with Mutual Information Neural Estimation for Text Generation,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (EMNLP-IJCNLP’19), November 2019, Hong Kong, China, pp. 4047–4057
- L. Song, C.W. Cheong, K. Yin, W.K. Cheung, B.C.M. Fung, J. Poon, “Medical Concept Embedding with Multiple Ontological Representations,” in *Proceedings of 2019 International Joint Conference on Artificial Intelligence*, Aug 10-16, 2019, Macao, China, pp. 4613-4619
- K. Yin, D. Qian, W.K. Cheung, B.C.M. Fung, J. Poon, “Learning Phenotypes and Dynamic Patient Representations via RNN Regularized Collective Non-negative Tensor Factorization,” in *Proceedings of the*

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- C. Li, W.K. Cheung, J. Liu, J.K. Ng. Automatic Extraction of Behavioral Patterns for Elderly Mobility and Daily Routine Analysis. *ACM Transactions on Intelligent Systems and Technologies*, 9(5): 54:1-54:26, 2018
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- Y. Zhang, W.K. Cheung and J. Liu, “A Unified Framework for Epidemic Prediction based on Poisson Regression,” *IEEE Transactions on Knowledge and Data Engineering*, Vol. 27. No. 11, pp.2878-2892, 2015
- J. Zeng, W.K. Cheung, J. Liu, "Learning Topic Models by Belief Propagation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(5), May 2013, 1121-1134